

Avijit Roy

Substitute Lecturer in Computer Science, John Jay College of Criminal Justice, CUNY

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RESEARCH INTERESTS

Security and robustness of machine learning systems; adversarial evaluation of perceptual and learned representations; security and reliability of LLM-based and agentic systems; multilingual and low-resource language technology, with emphasis on tokenization, representation, and the computational cost of access for speakers of underserved languages.

EDUCATION

M.S., Digital Forensics and Cybersecurity 2024 – 2026

John Jay College of Criminal Justice, The City University of New York

New York, NY

- Thesis: *Adversarial Robustness of Perceptual Hashing Systems: A Unified Security Evaluation Framework*. Advisor: Shweta Jain, Ph.D. ProQuest 3348071637.
- Claude Hawley Medal for the best master's thesis in the graduate program (2026).
- Degree awarded with Distinction. GPA: 3.88/4.00.

Advanced Certificate, Applied Digital Forensic Science

2024 – 2025

John Jay College of Criminal Justice, The City University of New York

New York, NY

- GPA: 4.00/4.00.

B.S., Computer Science and Information Security

2019 – 2021

John Jay College of Criminal Justice, The City University of New York

New York, NY

- Honors in Computer Science and Information Security. GPA: 3.96/4.00. Dean's List: Fall 2019, Spring 2020, Spring 2021.
- Leonard E. Reisman Medal for distinguished scholarship and service to the College (2021).
- Capstone research: machine-learning classification for cyber threat intelligence [R1]; vulnerability assessment of a WordPress deployment with OWASP ZAP and Burp Suite.

Undergraduate coursework (non-degree)

2011 – 2012

North South University

Dhaka, Bangladesh

- Credits transferred to the B.S. at John Jay College of Criminal Justice, CUNY.

PUBLICATIONS

Reverse chronological within each category. My name is in bold. Author order is as published.

Journal Articles

- [J1] Sonal Sagar Boda, Siddhesh Pimpale, Mohammad Somon Sikder, Hemayet Uddin Himel, Zeeshan Akbar, **Avijit Roy**, and Deepak Gupta. 2026. AI-Driven and Cyber-Secure Embedded Inverter Control for Energy Optimization in Electric Vehicles. *Applied Research* (Wiley) 5, 4 (2026), Article e70128. Special Issue on Next-Generation Artificial Intelligence and Security for Engineering Applications. doi:10.1002/appl.70128

Conference and Workshop Papers (peer-reviewed)

- [C1] **Avijit Roy**, Proma Roy, and Hrishitva Patel. 2026. Measuring the Tokenization Premium: A Cost Audit for Underserved Language Communities. In *Workshop on Language Models for*

Underserved Communities (LM4UC), co-located with the 35th International Joint Conference on Artificial Intelligence (IJCAI 2026). Bremen, Germany. arXiv:2608.09046 [cs.CL]
Oral and poster presentation. Benchmark and code: github.com/HeyAvijitRoy/tea-benchmark. Non-archival workshop.

- [C2] Udoy Sankar Saha, Hasan Imam, Md Mustafizur, **Avijit Roy**, Abdullah Al Zaiem, and Mia Md. Tofayel Gonee Manik. 2026. Deep Learning Based Malware Image Classification Using Binary Image Representation for Secure Management Information Systems. In *Proceedings of the 2026 Technology and Innovation Conference (TIC)*. IEEE, Kuala Lumpur, Malaysia, 1–6. doi:10.1109/TIC68483.2026.11702958
- [C3] Thomas Kimmeth, **Avijit Roy**, and Vivek Sharma. 2026. Propasafe-Hybrid: A Text-Based Hybrid Propaganda Detection Tool. In *Proceedings of the 39th International Florida Artificial Intelligence Research Society Conference (FLAIRS-39)*, Vol. 39, No. 1, Special Track on Applied Natural Language Processing. Marco Island, FL, USA. doi:10.32473/flairs.39.1.141595
Oral presentation at FLAIRS-39 and ILA 2026.
- [C4] **Avijit Roy**, Emran Hossain, Hemayet Uddin Himel, Hasan Imam, Ruhul Amin Md Rashed, and Harleen Kaur. 2026. Privacy-Preserving MIS Using Blockchain, Federated Learning and Zero-Trust Models. In *Innovative Computing and Communications: Proceedings of ICICC 2026* (Lecture Notes in Networks and Systems). Springer, Cham, 210–223. doi:10.1007/978-3-032-30008-9_16
Best Paper Award, 9th International Conference on Innovative Computing and Communication (ICICC 2026), New Delhi, India.

PREPRINTS AND WORK IN PROGRESS

Preprints

- [P1] **Avijit Roy** and Proma Roy. 2026. Structural Silence: When AI Infrastructure Fails Speakers of Underrepresented Languages — A Case Study in Bengali, the Digital Divide, and the Hidden Cost of English-Centric Design. Preprint. arXiv:2608.12278 [cs.CL]
An earlier version was posted to SSRN in April 2026 under the title “Structural Silence: When AI Infrastructure Fails the World’s Underrepresented Languages” (doi:10.2139/ssrn.6522858). Poster version presented at ILA 2026.

Work in Progress

- Bengali programming-assistance corpus and parameter-efficient fine-tuning for low-resource programming education (NSF ACCESS CIS260616). Data curation and training pipeline in progress.

Patent Applications

- [Pat1] **Avijit Roy**. 2026. Staged Client-Side Phishing Risk Assessment Architecture Using Locally Executed ONNX Transformer Inference Within an Email Client Runtime. U.S. Provisional Patent Application No. 63/986,762, filed February 2026.
Provisional application; not examined. Sole inventor.

Other Writing (not peer-reviewed)

- [R1] **Avijit Roy**. 2021 (archived 2026). Exploring Machine Learning Techniques for Cyber Threat Intelligence Analysis. Undergraduate research report, John Jay College of Criminal Justice, CUNY. doi:10.5281/zenodo.19056775

- [R2] **Avijit Roy**. 2026. *Cybersecurity Fundamentals: A Practical Guide for Everyone*. AP R&R Press. Kindle edition. General-audience book.
- [R3] **Avijit Roy**. 2026. *Saving Your Work Like a Pro: Git & GitHub for C++ Students*. Open educational e-book, 17 chapters. avijitroy.com/docs/git-version-control

RESEARCH EXPERIENCE

Graduate Researcher, Master's Thesis

Jan 2026 – Jun 2026

John Jay College of Criminal Justice, CUNY · Advisor: Shweta Jain

New York, NY

- Studied whether threshold-based perceptual hashing, which is used for image matching, content moderation, provenance, and forensic triage, stays reliable when an adversary optimizes perturbations against it.
- Built a matched evaluation protocol on a 500-image ImageNette validation subset for two attack goals: evasion, which moves a known image outside the match threshold, and near-collision, which moves an unrelated image inside it. Applied both to six configurations: pHash-64, pHash-256, PDQ, NeuralHash, and SmartHash in whole-image and overlapping-block variants. Exact collision was also tested for pHash-64 and NeuralHash, the two algorithms with prior exact-match baselines.
- Ran more than 40 attack runs of 500 images each on Jetstream2 GPU and large-memory nodes under NSF ACCESS allocation CIS260131.
- Found that all six configurations could be evaded for every tested image at operational thresholds while preserving high structural similarity. For pHash-64, collision success rose from 42.2% under exact matching to 99.4% under the threshold-based criterion ($T = 0.10$), so exact-match tests understate false-positive risk. Robustness depended more on pipeline structure, optimization landscape, and quantization design than on hash length; SmartHash's adaptive quantization reduced near-collision success under the tested surrogate.

Allocation Manager, NSF ACCESS Allocation CIS260131

Jan 2026 – Present

John Jay College of Criminal Justice, CUNY · PI: Shweta Jain

New York, NY

- Manage the allocation's Jetstream2 GPU and large-memory resources and run its experiments, including the master's thesis attack runs.

Principal Investigator, NSF ACCESS Explore Allocation CIS260616

Apr 2026 – Present

John Jay College of Criminal Justice, CUNY

New York, NY

- Project: *Dataset Development and Fine-Tuning of Language Models for Low-Resource Bengali Programming Assistance* (Deshi Mentor). Examines how data scarcity, tokenization overhead, and robustness gaps limit LLM programming help for Bengali-speaking learners.
- Built a validated corpus of 10,000 Bengali programming-assistance records (debugging tasks with explanations); curation is ongoing.
- Generated more than 8,000 multilingual adversarial test cases to compare model behavior on non-English adversarial input.
- Set up staged, versioned LoRA and QLoRA training and evaluation so that runs can be reproduced and compared across checkpoints.
- Designed the Tokenization Equity Audit (TEA) benchmark [C1]: a 120-item Python debugging corpus with semantically equivalent versions in English, Bengali, Hindi, Arabic, Tamil, and Yoruba, measured with the GPT-4o, Qwen2.5-7B, and Mistral-7B tokenizers. Bengali required $1.56\times$ the GPT-4o tokens of English and Yoruba $2.37\times$ despite its Latin script, so script family alone does not explain the premium.

NLP Research Contributor, Propasafe-Hybrid

Aug 2025 – May 2026

John Jay College, CUNY · NAIRR Pilot allocation 240383

New York, NY

- Contributed to a sentence-level propaganda-detection system that pairs a locally fine-tuned BERT classifier with LLM-based technique labeling and explanation, using cost-aware filtering to limit LLM calls [C3].
- Built ArticleExtractor, the ingestion component that strips advertisements, navigation, and dynamic page elements from news pages and outputs structured text for transformer and LLM analysis. Released as open source.
- Tested and debugged the system's Chrome browser extension.

Research Assistant and Platform Developer, LaPEL AI Strand

Jun 2025 – Present

John Jay College, CUNY · PI: Cristina Lozano-Argüelles

New York, NY

Paid Research Foundation of CUNY appointments Jun–Aug 2025 and Aug 2026; volunteer research collaborator between appointments.

- Studied whether LLM scoring can reproduce human ACTFL proficiency ratings of Spanish writing, mainly from heritage speakers, closely enough to support course placement. Part of LaPEL (Language Placement for Equitable Learning), a U.S. Department of Education–funded partnership between John Jay College and ILETC at the CUNY Graduate Center.
- Prepared 300 ACTFL-labeled writing samples; compared fine-tuned scoring with rubric-based few-shot prompting under identical input formats, and benchmarked commercial models against human-rated samples to select the model for the pilot.
- Measured agreement with human raters through confusion matrices, exact and adjacent-level accuracy, severe-error rates, and label coverage, which became the acceptance criteria for deployment. In the initial evaluation, the model matched the human rating exactly on 55% of 60 assessments and fell within one ACTFL sub-level on all 60; in a second validation run, the model selected for the pilot (OpenAI GPT-5.6 Terra) agreed with human ratings within one sub-level in every evaluated case.
- Sole developer of the placement platform, now in a live placement pilot at John Jay (React, FastAPI, PostgreSQL, AWS). It has three parts: a timed four-prompt writing assessment for students; a two-stage pipeline that separates the AI proficiency rating from version-controlled course-placement rules, so either can be evaluated or revised independently; and a staff portal for test administration, review, and analytics.
- Designed the privacy controls: student identity is stored encrypted and never sent to the scoring model, and research consent is recorded separately from access to placement, so students can be placed without joining the study.
- Named developer on a CUNY AI Innovation Fund proposal (submitted September 2026) to validate the platform against two ACTFL-certified raters on writing samples from 70–100 students, refine it from observed use, and run a full placement cycle.

Capstone Researcher; Graduate Research Assistant from Jan 2025

Aug 2024 – May 2025

John Jay College, CUNY · Supervisor: Shweta Jain

New York, NY

- Recovered a non-functional multi-agency platform for triaging human-trafficking tips, built for an industry partner, and migrated its backend from .NET 5 to .NET 9.
- Designed the security layer the system had never had: Azure AD OAuth 2.0, TLS in transit, field-level encryption at rest, role-based access control, and audit logging with access-pattern anomaly detection.
- Replaced a paid ArcGIS mapping service with a real-time geospatial layer (Google Maps API, SignalR, Hangfire) and deployed the system to Microsoft Azure.

TEACHING EXPERIENCE

Adjunct Assistant Professor

Sep 2026 – Present

Collins College of Professional Studies, St. John's University

Queens, NY

- DFR 711, *Cyber-Forensic and Malware Analysis* (graduate), Fall 2026. Rebuilt the syllabus, lectures, and lab sequence for the M.S. in Cyber and Information Security: incident response, evidence acquisition, memory and storage analysis, static and dynamic malware analysis, and forensic reporting on Windows, macOS, and Linux.

Substitute Lecturer (full-time)

Aug 2026 – Present

Department of Mathematics and Computer Science, John Jay College, CUNY

New York, NY

- Fall 2026, instructor of record, 28–30 students per section: CSCI 272 *Object-Oriented Programming in C++* (two sections); CSCI 362 *Databases and Data Mining*; CSCI 377 *Computer Algorithms*.

Adjunct Lecturer

Aug 2025 – Aug 2026

Department of Mathematics and Computer Science, John Jay College, CUNY

New York, NY

- CSCI 271 *Introduction to Computing and Programming* (Spring 2026); CSCI 272 *Object-Oriented Programming* (Fall 2025, Spring 2026).
- Developed the curriculum for CSCI 380 *Agentic AI Systems*, a selected-topics course on LLM workflows, tool use, retrieval-augmented systems, multi-agent design, and evaluation (approved concept; awaiting enrollment).
- Rebuilt the CSCI 271 lecture notes, slides, and course content so that the course prepares students for the object-oriented material in CSCI 272.
- Wrote open course materials: a 17-chapter Git e-book, C++ toolchain guides for Windows and macOS, and a guide to responsible AI use in coursework.

Research Mentor, Summer Undergraduate Research Program

Jun – Aug 2025

Department of Mathematics and Computer Science, John Jay College, CUNY

New York, NY

- Mentored five undergraduate researchers building a Django/PostgreSQL authentication system that enforces one active session per user; led design and code reviews and brought the project to a public, reproducible repository with tests and deployment documentation.

Lecturer in Computer Science

Jul 2018 – Jul 2019

United College of Aviation Science and Management

Dhaka, Bangladesh

- Taught introductory computing and programming to about 50 students in two sections.

AWARDS, FELLOWSHIPS, AND RESEARCH SUPPORT

Awards and Honors

Claude Hawley Medal, best master's thesis, John Jay College, CUNY	2026
Graduate Achievement Award, John Jay College, CUNY	2026
Best Paper Award, ICICC 2026, for [C4]	2026
Student Council Scholarship, John Jay College, CUNY	2026
Leonard E. Reisman Medal, outstanding member of the senior class, John Jay College	2021
Deserve Scholarship (\$1,000; 50 recipients nationally)	2019
CUNY travel funding, Student Conference on U.S. Affairs (SCUSA 71), U.S. Military Academy	2019

Research Computing Support

NSF ACCESS Explore allocation CIS260616 (PI): 120,000 SUs Jetstream2 GPU, 50,000 SUs Jetstream2 Large Memory, 30,000 ACCESS credits (estimated value \$31,305) 2026 – 2027
NSF ACCESS allocation CIS260131 (allocation manager; PI: Shweta Jain) 2026 – Present
NAIRR Pilot allocation 240383 (contributor, Propasafe-Hybrid) 2025 – 2026
John Jay Student Travel Fund award, travel to FLAIRS-39 2026

Grant Proposals (pending)

CUNY AI Innovation Fund, LaPEL AI strand: validation and deployment of an AI-assisted Spanish placement platform. Role: developer (PI: C. Lozano-Argüelles; co-director: A. Gatti, ILETC). Total request \$25,000. Submitted September 29, 2026.

PRESENTATIONS AND POSTERS

- *Measuring the Tokenization Premium: A Cost Audit for Underserved Language Communities* [C1]. Oral and poster presentation, LM4UC Workshop at IJCAI 2026, Bremen, Germany. August 16, 2026.
- *Propasafe-Hybrid: A Text-Based Hybrid Propaganda Detection Tool* [C3]. Oral presentation, FLAIRS-39, Special Track on Applied NLP, Marco Island, FL. May 18, 2026.
- *Fine-Grained Sentence-Level Propaganda Detection in News Articles*, by M. Eren, A. Shakeel, and V. Sharma. Presented on behalf of the authors (not an author), FLAIRS-39, Special Track on Applied NLP, Marco Island, FL. May 18, 2026.
- *Adversarial Robustness of Perceptual Hashing Systems: A Unified Security Evaluation Framework*. Graduate Student Research Symposium, John Jay Research and Creativity Expo, New York, NY. May 4, 2026.
- *Propasafe-Hybrid* [C3]. Oral presentation, 69th Annual Conference of the International Linguistic Association (ILA 2026), New York, NY. April 30–May 2, 2026.
- *Structural Silence* [P1]. Poster, ILA 2026, New York, NY. April 30–May 2, 2026.

MEDIA

“Avijit Roy ’21, ’25, ’26: Computer Scientist Protecting People in the Age of AI.” Alumni profile, John Jay College News. jjay.cuny.edu Jun 2026

PROFESSIONAL SERVICE AND LEADERSHIP

Academic Service

Reviewer, *Information Systems Frontiers* (Springer) Oct 2026
Judge, HackKnight 2026 student hackathon, John Jay College Oct 2026
Moderator, alumni panel, CUNY International Student Success Conference Sep 2026
Organizer and planner, CS OOP Hackathon and Capture-the-Flag (CS Department and CSS collaboration), John Jay College May 2026

University Governance

Graduate Representative, John Jay College Council; Committee on Graduate Studies; 2024 – 2025
Faculty–Student Disciplinary Committee
International Representative, John Jay Student Council 2020 – 2021
Alternate Delegate, CUNY University Student Senate 2020 – 2021
Vice President (2020–21) and Secretary (2019–20), International Students Association, 2019 – 2021
John Jay

Community and Volunteer Service

Contributor, OpenStreetMap 2015 – Present
Global Guide, One To World (classroom programs for New York City students) 2019 – 2021
Volunteer photographer, CITYarts, Inc. 2016 – 2018
Executive Manager, Mapping Bangladesh community (Dhaka and Noakhali districts) 2013 – 2017
Citizen Cartographer (2009–12), Regional Expert Reviewer (2012–15), and Regional 2009 – 2016
Lead (2015–16), Google Map Maker; invited participant, Google SEArch Summit,
Yogyakarta, Indonesia (2013)

OTHER PROFESSIONAL EXPERIENCE

Data Analyst (part-time Non-Teaching Adjunct) Jul 2024 – Present
International Student and Scholar Services, CUNY Central Office New York, NY

- Validate more than 10,000 SEVIS records per term, each with 60+ attributes, against CUNYfirst, CUNY's system of record for more than 225,000 students, across the 25 CUNY colleges that enroll international students; reconcile records across relational tables and flag discrepancies for campus follow-up (SQL, Python).
- Translate federal SEVP reporting rules and institutional policy into explicit validation checks and Power Automate flows that support CUNY's biennial SEVP recertification.
- Produce term-over-term trend analyses and enrollment projections from historical SEVIS and enrollment data for university planning; develop and maintain Power BI, Tableau, and Excel dashboards and documented data-governance procedures.

University Student Records Coordinator (full-time) Jun 2023 – May 2024
International Student and Scholar Services, CUNY Central Office New York, NY

- Built Excelsior, a C#/.NET tool that matches student records across files by unique identifier and indexes discrepancies; each run compares 1,200–7,000 records, and the tool cut manual verification time by about 30%.
- Translated regulatory and institutional rules into validation checks applied consistently by the tool, and automated audit-ready exception summaries (Excel/VBA) for the Registrar and ISSS teams.

IT Specialist and Data Analyst (College Assistant) Oct 2019 – Jun 2023
International Student and Scholar Services, CUNY Central Office New York, NY

- Built data-collection pipelines for F-1 student data across all CUNY campuses, reducing collection time from 30 days to 2.
- Wrote SQL queries and stored procedures to reconcile and analyze 10,000+ student records across relational systems; cleaned and migrated legacy records and added data-quality checks.
- Built standardized Tableau and Excel reporting templates (about 80% faster reporting) and ran cohort-level enrollment and retention analyses for planning.

Technology and Learner Engagement Specialist, New York Jobs CEO Council Aug 2021 – May 2022

Chief Technology Officer, Carpet Culture (remote)	Apr 2018 – Jul 2019
Senior Web Engineer and Sales Executive, Carpet Culture	Mar 2017 – Mar 2018
Web Developer Intern, Carpet Culture, New York, NY	May – Aug 2016
IT Officer and Business Development Manager (part-time), United College of Aviation Science and Management, Dhaka. Built the campus network and computer labs; designed the admissions counseling process.	Jun 2011 – Aug 2013
Debate Panelist (freelance), weekly social-awareness programs for the Population and Health Department, Bangladesh Betar (national public radio), Dhaka	Oct 2011 – Aug 2013
System Engineer (part-time), Techmania, Dhaka	Jun – Dec 2010

TECHNICAL SKILLS

Programming	Python, C++ , C#, SQL, JavaScript/TypeScript, VBA
ML and NLP	Hugging Face Transformers, BERT fine-tuning, LoRA/QLoRA, ONNX inference, LLM APIs (OpenAI, Anthropic, Google)
Security and forensics	Autopsy, FTK Imager, Wireshark/TShark, HxD, Burp Suite, OWASP ZAP, MITRE ATT&CK
Systems	Linux (Ubuntu, Kali) and command-line tooling; .NET 9/ASP.NET Core, FastAPI, Django, React, PostgreSQL, MySQL, Docker, GitHub Actions
Cloud	AWS (CDK, ECS Fargate, RDS), Microsoft Azure, Google Cloud
Research computing	NSF ACCESS Jetstream2 (GPU and large-memory), reproducible versioned experiment pipelines, $\LaTeX$
Training	Certified Peer Reviewer Course, Elsevier Researcher Academy (2026); Google Cybersecurity Professional Certificate (2024)
Languages	Bengali (native), English (fluent), Hindi (spoken)